Model Card: Home Credit Payment-Difficulty Model

XGBoost credit-risk ranking model — performance, economics, explainability, fairness and deployment readiness

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> **Status: research baseline — not approved for automated lending.** TARGET = 1 means *payment difficulty* (a late installment early in the loan), not a confirmed default or charge-off. All money figures are hypothetical scenario estimates, not observed lender profit.

1 Executive Summary

Business problem. Home Credit lends to people with little or no conventional credit history. This model estimates each applicant's probability of payment difficulty so that applications can be ranked by risk, which would let the lender focus scrutiny on the riskiest applicants without turning away borrowers who can repay.

What we found. On 61,502 labeled applicants that were held back and never used for training, tuning, calibration or threshold selection:

Table 1: Headline results on the untouched holdout (frozen 0.25 research threshold)

Measure	Result
Ranking quality (ROC AUC)	0.7551 [0.7481, 0.7622]
Approval rate at 0.25	94.9% [94.8%, 95.1%]
Precision (declined who had payment difficulty)	33.6% [31.9%, 35.2%]
Recall (payment-difficulty cases declined)	21.1% [20.0%, 22.2%]
Gain vs approving everyone — hypothetical 24-month scenario	$15.84 per applicant [$12.86, $19.07]
Gain vs approving everyone — hypothetical 36-month scenario	$0.79 per applicant [−$2.62, $4.46] — spans zero

The model ranks risk consistently: holdout AUC 0.7551 against 0.7592 in development cross-validation. Its calibrated probabilities match observed rates overall and within gender and education groups. The **0.25 threshold was fixed from development data before the holdout was opened.** At that threshold about one applicant in twenty is declined or referred, and the payment-difficulty rate among approved applicants falls from 8.07% (all holdout applicants) to 6.71%.

Recommendation: keep this model as a research baseline. Do not use it for automated approvals or denials. Four problems each block deployment on their own:

1. **It uses gender and marital status directly.** Gender is its fourth most influential input. About two-thirds of the 3.9-point approval gap between women and men comes from that direct use. Both are prohibited bases under U.S. fair-lending rules (Regulation B).
2. **It uses age in a way that sometimes counts against applicants aged 62 and over**, which the age rules for credit scoring do not allow.
3. **Its main driver is a set of external scores whose source and meaning are undocumented**, so it cannot give declined applicants specific, accurate reasons.
4. **Its financial case is hypothetical.** Under a plausible 36-month loan term the threshold adds essentially nothing, and the label measures early payment difficulty, not losses.

Before deployment could be considered:

- legal review of every restricted or potentially discriminatory input;
- a rebuilt, compliant candidate model validated the same way;
- documentation of the external scores and tested adverse-action reason codes;
- validation against real loss, term and cost data;
- independent model validation with ongoing fairness and drift monitoring.

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2 Model Details

Table 2: Model facts

Item	Detail
Model type	Gradient-boosted decision trees (XGBoost), binary classification, `binary:logistic` objective
Software	R version 4.5.2 (2025-10-31 ucrt); xgboost 3.2.1.1; caret 7.0.1 (custom caret method, because caret's built-in `xgbTree` does not work with xgboost 3.x)
Final saved model	`models/final_home_credit_xgb.rds` : caret `train` object fit on all 307,511 labeled rows; produced the Kaggle submission
Training data (final model)	307,511 applicants from Kaggle `application_train.csv` ; 24,825 with TARGET = 1 (8.07%)
Target	TARGET = 1: "late payment more than X days on at least one of the first Y installments" (X and Y not disclosed). TARGET = 0: all other cases.
Predictors	167 engineered predictors → 300 columns after dummy encoding; 71 columns are actually used by the depth-2 trees
Hyperparameters	nrounds = 300, max_depth = 2, eta = 0.047, gamma = 4, colsample_bytree = 0.95, min_child_weight = 10, subsample = 0.86; eval_metric = logloss; nthread = 4
Class-imbalance handling	None (weighting and downsampling both lowered cross-validated AUC)
Calibration (model card)	Platt scaling fitted on development out-of-fold predictions: p = logistic(0.1903 + 1.0905 × logit(raw p))
Decision rule (research)	Approve if calibrated probability < 0.25; otherwise decline or refer

Data sources. The data are Kaggle's *Home Credit Default Risk* competition files. The model uses `application_train.csv` (one row per applicant) and `bureau.csv` (1.7 million prior credits reported by other lenders), aggregated to one row per

applicant. The other supplementary tables (`previous_application`, `installments_payments`, `POS_CASH_balance`, `credit_card_balance`, `bureau_balance`) were not used. The data's country, currency and time period are not disclosed.

Feature engineering (`data_preparation.R`, driven by decisions recorded in the EDA):

- **Cleaning:** income capped at the training 99th percentile; the 365243 "days employed" sentinel recoded to a not-employed flag; rare categories pooled.
- **Affordability ratios:** loan-to-value, payment-to-income, credit-to-income, credit-to-annuity.
- **External-score summaries:** mean, minimum and maximum.
- **Tenure and age** expressed in years.
- **Bureau aggregates:** open credits, debt-to-credit, overdue amounts, history length.
- **Missing-value handling:** 18 missing-value flags, plus bands and interaction terms.
- **Train/test consistency:** every learned parameter comes from training data and is re-applied unchanged to test data.

How the model was built and validated.

1. **Selection** (`modeling.qmd`): a stratified 80/20 development/holdout split (seed 6493). Logistic regression, random forest and XGBoost were compared by 3-fold CV AUC; XGBoost won. Imbalance strategies were tested, and a 20-configuration randomized search was run on development data.
2. **Model-card validation** (this document):
    - five-fold cross-validation on the 246,009 development rows, with preprocessing re-learned inside each fold;
    - calibration and threshold choice from development data only;
    - one evaluation of a development-only model on the untouched 61,502-row holdout.

Reproducibility. Reloading the saved model reproduces all 48,744 Kaggle submission predictions; the largest difference is 5.6e-17, which is floating-point noise from the CSV round-trip, and all 12 checks passed. The development-only model reproduces the notebook's historical holdout AUC of 0.7556 exactly when the original preprocessing is used. Analysis scripts and saved outputs are in `models/model_card/` (git-ignored because of size and data licensing). This notebook reads those outputs and does not retrain anything.

3 Intended Use

Table 3: Intended and unsupported uses

Aspect	Description
Primary intended use	Research and education: ranking loan applicants by estimated risk of early payment difficulty, and studying the trade-offs between performance, economics, explainability and fairness.
Possible future decision support (only after the conditions in Section 9 are met)	Prioritizing applications for human underwriting review; routing higher-risk applications to extra income or document verification; informing portfolio-level risk monitoring.
Intended audience	Credit-risk and analytics teams, model-risk management, compliance and legal reviewers, and business leadership evaluating whether a compliant version is worth building.
Out of scope / not supported	**Automated approval or denial of credit**; setting individual prices, limits or terms; generating adverse-action notices; use in any market, product or period other than the one the data represent; any use while gender, marital status or unrestricted age remain inputs.

4 Performance Metrics

Three kinds of results appear below. They come from different data and models, so they should not be ranked against each other:

- **New model-card validation:** five-fold development cross-validation and the untouched holdout.
- **Historical results** from `modeling.qmd`.
- **Kaggle scores**, recorded by hand.

Table 4: ROC AUC across evaluations

Evaluation	Model	ROC AUC	Note
Five-fold cross-validation (new)	Final configuration, 5 × ~197k-row fits, fold-specific preprocessing	0.7592 ± 0.0011 (pooled 0.7592)	Development data only
Untouched holdout (new, primary)	Final configuration fit on 246,009 development rows, development-only preprocessing	0.7551 [0.7481, 0.7622]	Single, final evaluation
Historical holdout (`modeling.qmd`)	Same configuration, original whole-data preprocessing (reproduced exactly in this work)	0.7556	Recorded in the modeling notebook
Kaggle public score (historical)	Final model refit on all 307,511 rows	0.75749	Late submission; entered manually from Kaggle
Kaggle private score (historical)	Final model refit on all 307,511 rows	0.75276	Late submission; entered manually from Kaggle
Historical exploration CV (untuned configuration)	Depth-5 starting configuration, 50,001-row sample, 3-fold	0.7575	Different model settings; not comparable
Historical tuning CV	Best of 20 configurations, 5,000-row sample, 3-fold	0.7085	Much smaller sample; not comparable

The holdout AUC is 0.0041 below cross-validation, and the confidence intervals overlap. The holdout and both Kaggle scores agree within a few thousandths. Together, these suggest the model generalizes to new applicants drawn from the same population.

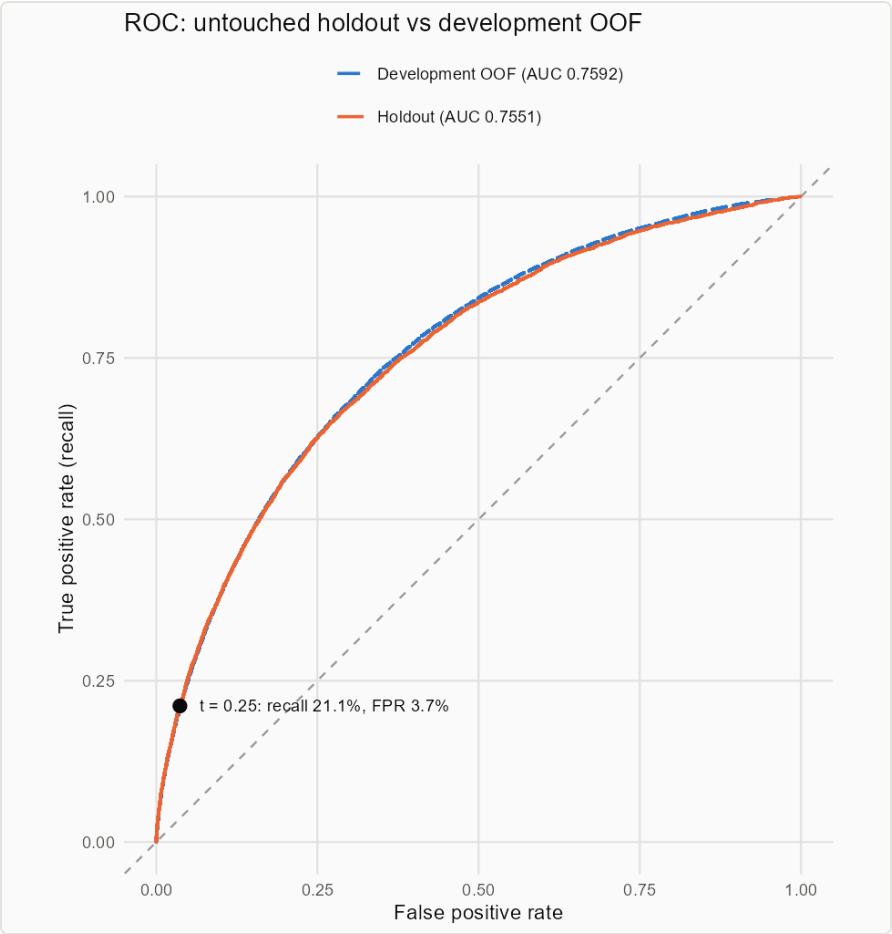


Figure 1: ROC curve on the untouched holdout (orange) and development out-of-fold predictions (blue, dashed). The dot marks the frozen 0.25 threshold.

Classification at the 0.25 research threshold. In this card the "positive" prediction is *decline or refer*. Precision is the share of declined applicants who actually had payment difficulty; recall is the share of payment-difficulty applicants who were declined.

Table 5: Holdout confusion matrix at threshold 0.25 (n = 61,502)

	Declined / referred (p ≥ 0.25)	Approved (p < 0.25)	Total
Actual: payment difficulty (TARGET = 1)	1,047	3,918	4,965
Actual: no payment difficulty (TARGET = 0)	2,073	54,464	56,537
Total	3,120	58,382	61,502

Table 6: Threshold metrics: holdout (with 95% Wilson intervals) vs development out-of-fold

Metric	Holdout	Development OOF
Approval rate	94.9% [94.8%, 95.1%]	94.8%
Decline / referral rate	5.1%	5.2%
Precision	33.6% [31.9%, 35.2%]	33.1%
Recall (true positive rate)	21.1% [20.0%, 22.2%]	21.4%
Specificity	96.3% [96.2%, 96.5%]	96.2%
False positive rate	3.7% [3.5%, 3.8%]	3.8%
Payment-difficulty rate among approved	6.71% [6.51%, 6.92%]	6.69%
Payment-difficulty rate among declined	33.6% [31.9%, 35.2%]	33.1%

Calibration. The raw XGBoost scores are slightly compressed (calibration slope about 1.08). The Platt correction, fitted only on development predictions, carries over to the holdout. Near the threshold the calibrated scores run about one point high (0.20–0.25 band: 22.3% predicted vs 21.3% observed).

Table 7: Probability accuracy on the holdout and in development cross-validation

Data	Score	Brier	Log loss	Mean predicted / observed	Calibration intercept [95% CI]	Calibration slope [95% CI]
Holdout	Raw	0.0678	0.2468	8.00% / 8.07%	0.010 [-0.020, 0.040]	1.081 [1.047, 1.116]
Holdout	Platt (frozen)	0.0678	0.2466	8.00% / 8.07%	0.011 [-0.019, 0.041]	0.991 [0.960, 1.023]
Development OOF	Raw	0.0678	0.2461	8.07% / 8.07%	-0.000 [-0.015, 0.015]	1.091 [1.073, 1.108]
Development OOF	Platt (cross-fitted)	0.0678	0.2459	8.07% / 8.07%	-0.000 [-0.015, 0.015]	1.000 [0.984, 1.016]

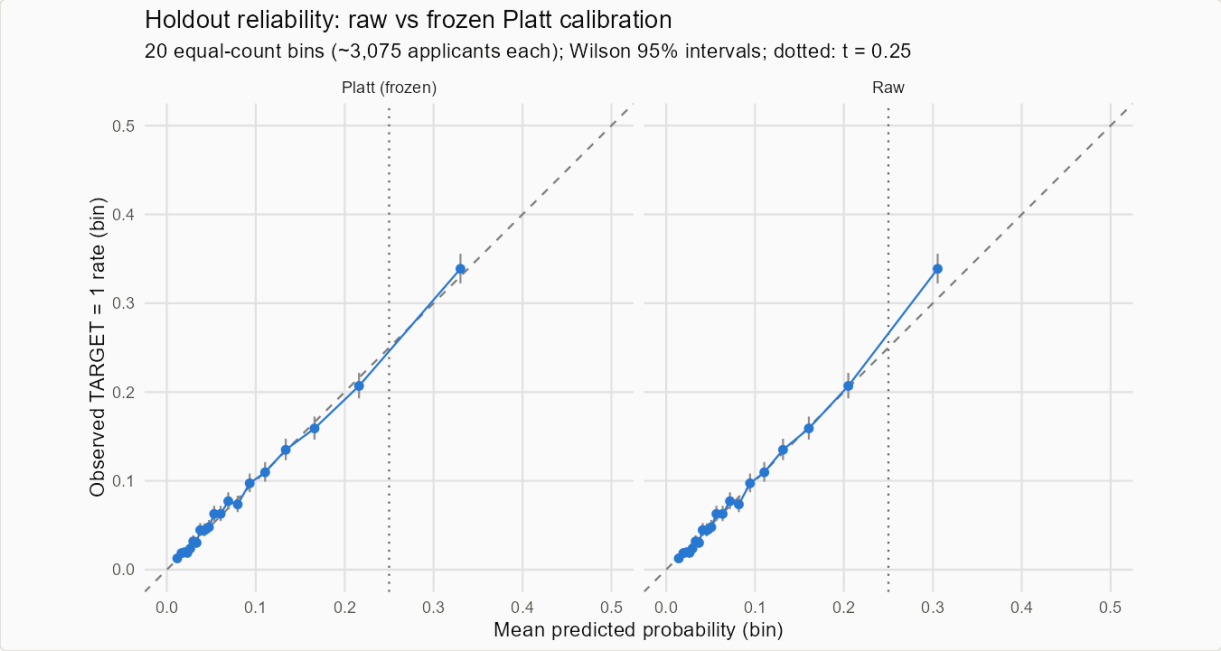


Figure 2: Holdout reliability: observed payment-difficulty rate against mean predicted probability in 20 equal-size bins, for raw and Platt-calibrated scores. The dashed diagonal is perfect calibration; bars are 95% intervals; the dotted line is the 0.25 threshold.

5 Decision Threshold Analysis

Hypothetical economics

Every money figure in this section comes from **scenario assumptions** applied to the model's predictions. None is an observed profit. TARGET = 1 is a payment difficulty, so turning it into a loss requires an assumed share of such cases that end in charge-off (*c*). No source exists for that share.

Method. Each approved applicant contributes the lifetime margin of a repaid loan, **G**, if they have no payment difficulty, and loses **L = c × EAD × LGD** if they do. Declined applicants contribute zero. Funding and operating costs sit only in G, and recoveries only in the loss given default (LGD), so nothing is counted twice. Expected profit is maximized by approving whenever an applicant's calibrated probability is below the **break-even probability**:

$$p^* = G / (G + L) \text{ approve when calibrated } p < p^*; \text{ decline or refer otherwise}$$

Sourced inputs. Home Credit's own FY2018 results give a net interest margin of 15.8% and a cost-to-income ratio of 42.2%. Approved prior cash loans in the dataset imply a median 24-month term at about 40% APR. Loss given default spans 45% (the Basel senior-unsecured benchmark) to 96% (debt buyers paid about 4 cents per dollar, FTC 2013). Contextual figures come from the Federal Reserve, FDIC and TransUnion (see Section 10). The dollar amounts use the TransUnion average personal-loan balance of $8,437 purely for illustration; the dataset's currency is undisclosed, and p* does not depend on loan size.

Table 8: Financial scenarios (per unit of principal; dollar values at P = $8,437 for illustration)

Scenario	Annual margin m	Profit if repaid, G	Exposure at default	LGD	Loss conversion c	Loss per TARGET = 1, L	Break-even p*	Margin source
Conservative	3.07%	0.036 ($303)	1.000	96.0%	100%	0.960 ($8,100)	0.036	FDIC consumer lenders: NIM + nonint inc - exp
Base	9.13%	0.107 ($902)	0.914	70.5%	50%	0.322 ($2,717)	0.249	Home Credit FY2018: NIM x (1 - CIR 42.2%)
Optimistic	9.57%	0.112 ($946)	0.818	45.0%	25%	0.092 ($777)	0.549	Home Credit Q1 2019: NIM x (1 - CIR 39.4%)

The base scenario combines Home Credit's margin, three installments paid before default, the midpoint loss given default (70.5%, an assumption) and *c* = 50% (an assumption). Its break-even probability is **0.2493**, which rounds to the **0.25 research threshold**.

Why 0.25 was chosen before the holdout was opened:

- It is the theoretical base-case break-even, rounded.
- The profit-maximizing threshold found empirically on development out-of-fold predictions was 0.2525, with a bootstrap interval of 0.2475–0.275.

- The base-case profit curve is flat between about 0.23 and 0.30.
- The threshold, the calibration and the financial assumptions were all frozen before any holdout outcome was examined.

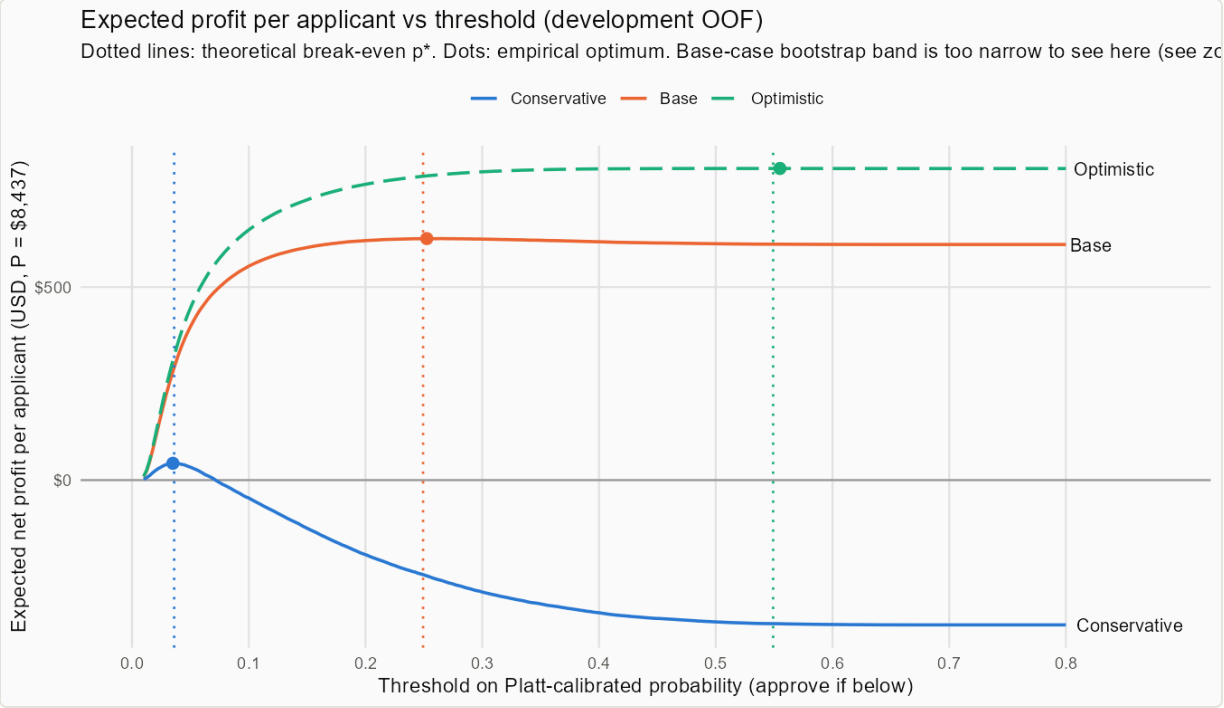


Figure 3: Development data: expected profit per applicant against threshold for the three scenarios (hypothetical). Dotted lines mark each scenario's theoretical break-even p*; dots mark the empirical optimum.

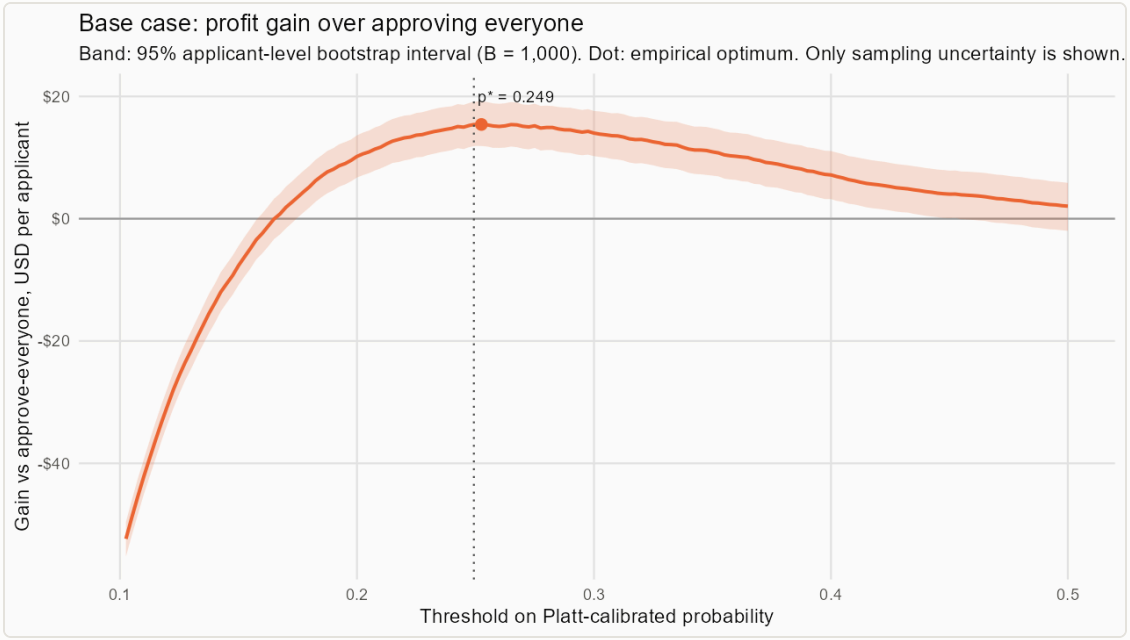


Figure 4: Development data, base scenario: gain over approving everyone, with a 95% bootstrap band. Only sampling uncertainty is shown, not assumption uncertainty.

Holdout outcomes at the frozen 0.25 threshold (61,502 applicants; 58,382 approved, 3,120 declined or referred):

Table 9: Hypothetical financial outcomes on the holdout at t = 0.25 (USD at P = $8,437; bootstrap 95% CI for the gain)

Scenario	Profit / applicant	Profit / approved loan	Approve-everyone / applicant	Gain vs approve-everyone / applicant	Total gain	Avoided proxy losses	Forgone margin
Base (24-month)	$626.09	$659.55	$610.25	**$15.84** [$12.86, $19.07]	$0.97M	$2.84M	$1.87M
Base, 36-month term	$1,067.82	$1,124.89	$1,067.04	**$0.79** [−$2.62, $4.46]	$0.05M	$2.97M	$2.92M
Conservative	−$247.32	−$260.54	−$374.98	**$127.66** [$119.83, $136.80]	$7.85M	$8.48M	$0.63M
Optimistic	$788.42	$830.55	$807.09	**−$18.67** [−$20.19, −$17.14]	−$1.15M	$0.81M	$1.96M

- **Base, 24-month:** the gain over approving everyone is **$15.84 per applicant**, or about $0.97M across the holdout. The development estimate was $15.39, so the result held up.
- **36-month term:** the gain falls to **$0.79**, with an interval spanning zero. Longer loans earn more margin while losses barely change, which moves the break-even point up to 0.332. The data hint that current loans may run about 36 months (median credit/annuity ratio 20.7), so this term uncertainty is unresolved and material.
- **Conservative:** the threshold cuts losses sharply, but the portfolio still loses money.
- **Optimistic:** 0.25 is too strict and gives up more margin than it saves.

Table 10: Base-case break-even probability by assumed loan term (40% APR)

Term (months)	Profit if repaid G	Loss L	Break-even p*
12	0.052	0.277	0.159
24	0.107	0.322	0.249
36	0.167	0.336	0.332
48	0.232	0.343	0.404

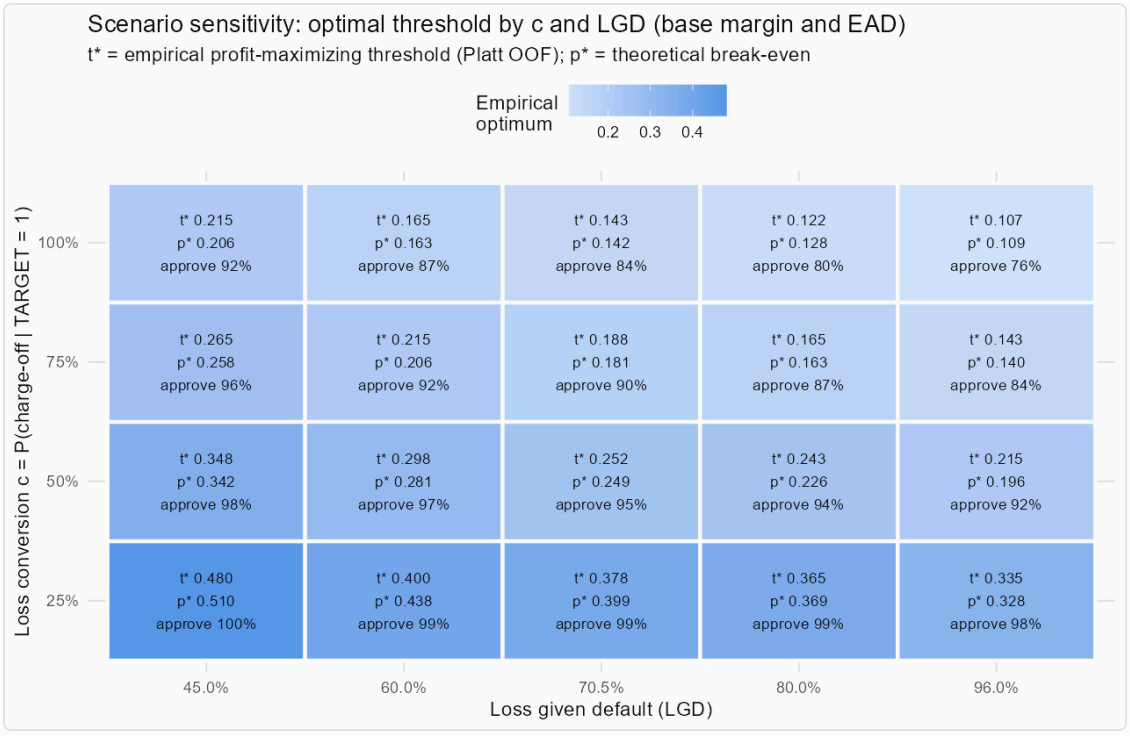


Figure 5: Development data: the optimal threshold depends mostly on the two least-supported assumptions, the loss-conversion rate c and loss given default. It ranges from about 0.11 to 0.48.

6 SHAP Explainability

Method.

- **Model explained:** the **original final model** trained on all 307,511 applicants, so these are *fitted-model* explanations rather than evidence of out-of-sample performance.
- **Sample:** a reproducible sample of **1,000 labeled applicants** (seed 20261025).
- **Algorithm:** xgboost's built-in exact TreeSHAP (Lundberg et al., 2020).
- **Units:** SHAP values are in **log-odds of payment difficulty**. For every applicant the base value (−2.442 log-odds, p = 0.080) plus the SHAP values equals the model's log-odds, within 4×10^{-6}. They are **not** percentage-point changes. Where this card quotes percentage points, they come from an explicit conversion.
- **Coverage:** 71 of the 300 encoded columns have any effect.

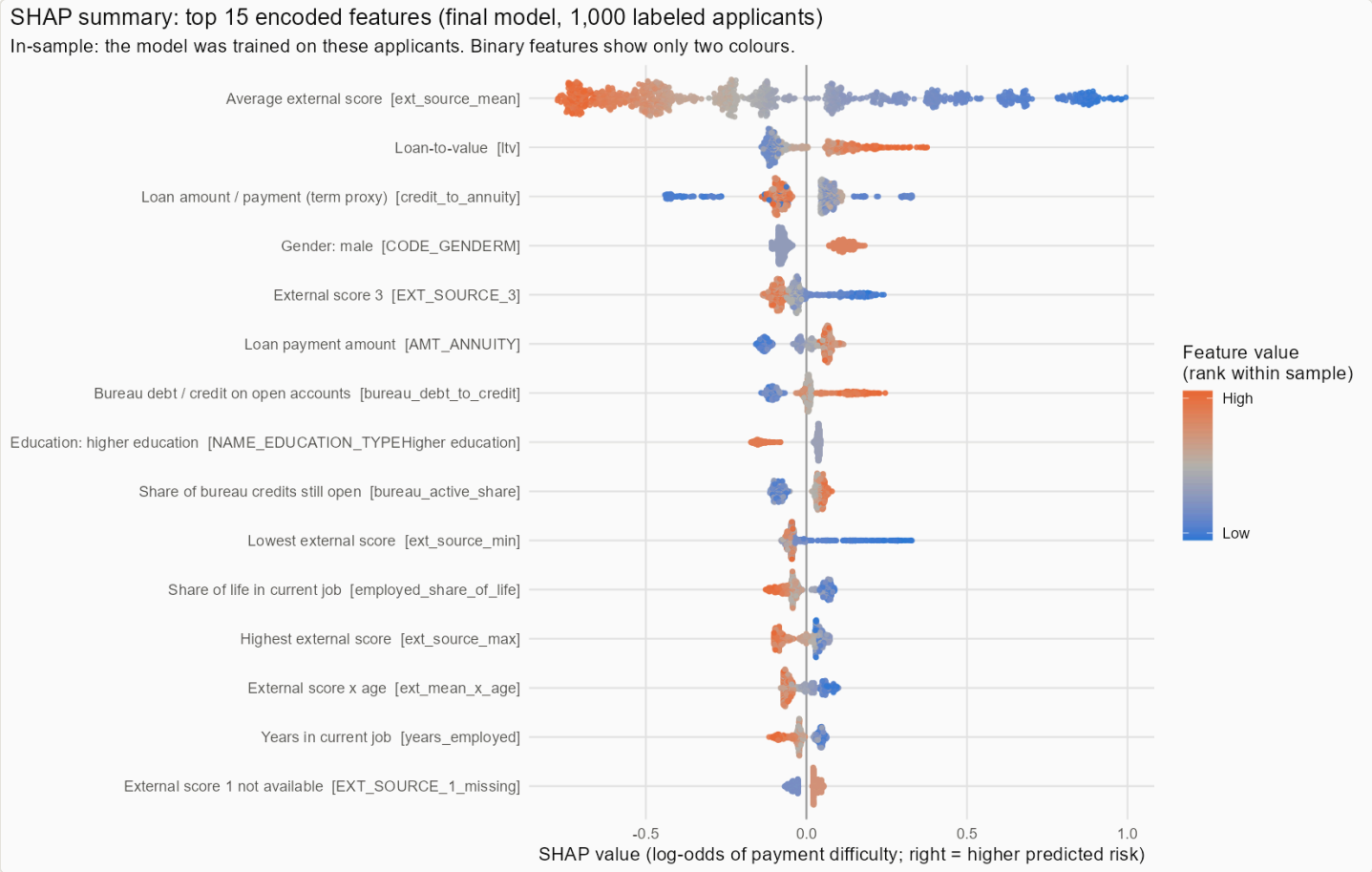


Figure 6: SHAP summary for the top 15 encoded features. Each dot is one applicant: position shows the push toward higher (right) or lower (left) predicted risk; color shows the feature's value (blue low, orange high).

Table 11: Top 15 predictors by mean absolute SHAP value (log-odds; in-sample, full-data model)

#	Feature	Meaning	Mean |SHAP|	Share	Direction
1	`ext_source_mean`	Average external score	0.442	22.7%	Higher values lower predicted risk
2	`ltv`	Loan-to-value	0.111	5.7%	Higher values raise predicted risk
3	`credit_to_annuity`	Loan amount / payment (term proxy)	0.107	5.5%	Non-monotonic / interaction-dependent
4	`CODE_GENDERM`	Gender: male	0.092	4.7%	Having this attribute raises predicted risk
5	`EXT_SOURCE_3`	External score 3	0.074	3.8%	Higher values lower predicted risk
6	`AMT_ANNUITY`	Loan payment amount	0.070	3.6%	Higher values raise predicted risk
7	`bureau_debt_to_credit`	Bureau debt / credit on open accounts	0.067	3.4%	Higher values raise predicted risk
8	`NAME_EDUCATION_TYPEHigher education`	Education: higher education	0.064	3.3%	Having this attribute lowers predicted risk
9	`bureau_active_share`	Share of bureau credits still open	0.061	3.1%	Higher values raise predicted risk
10	`ext_source_min`	Lowest external score	0.059	3.1%	Non-monotonic / interaction-dependent
11	`employed_share_of_life`	Share of life in current job	0.059	3.0%	Higher values lower predicted risk
12	`ext_source_max`	Highest external score	0.051	2.6%	Higher values lower predicted risk
13	`ext_mean_x_age`	External score x age	0.050	2.6%	Higher values lower predicted risk
14	`years_employed`	Years in current job	0.041	2.1%	Higher values lower predicted risk
15	`EXT_SOURCE_1_missing`	External score 1 not available	0.036	1.8%	Having this attribute raises predicted risk

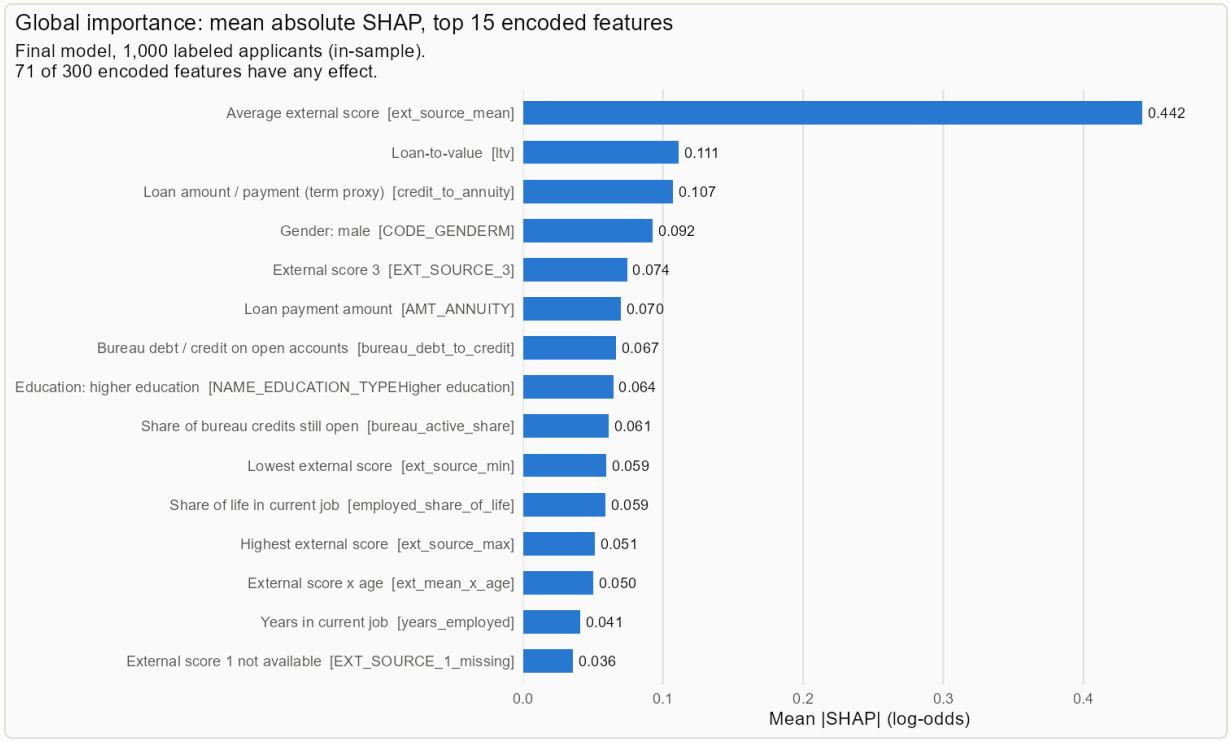


Figure 7: Global importance: mean absolute SHAP value for the top 15 encoded features.

What drives predictions.

- **External scores dominate.** The undocumented external-score family accounts for about half of all attribution (mean |SHAP| 0.60 log-odds as a family). Higher scores lower predicted risk.
- **Loan structure** comes next: borrowing more than the goods cost (loan-to-value above about 1.16), larger payments, and loans in the middle range of credit-to-payment ratios all raise risk.
- **Existing debt:** a high share of open bureau credit still owed, or a high share of bureau accounts still open, raises risk.
- **Job stability:** longer tenure lowers risk.
- **Protected and sensitive characteristics also matter.** Gender is the #4 feature (being male raises predicted risk) and higher education is #8 (it lowers risk).
- **Features with little direct effect:** `AMT_CREDIT` itself has none, plain payment-to-income is minor, and the direct age term ranks only #22 because its effect is spread across interaction features.

These are predictive associations, not causes.

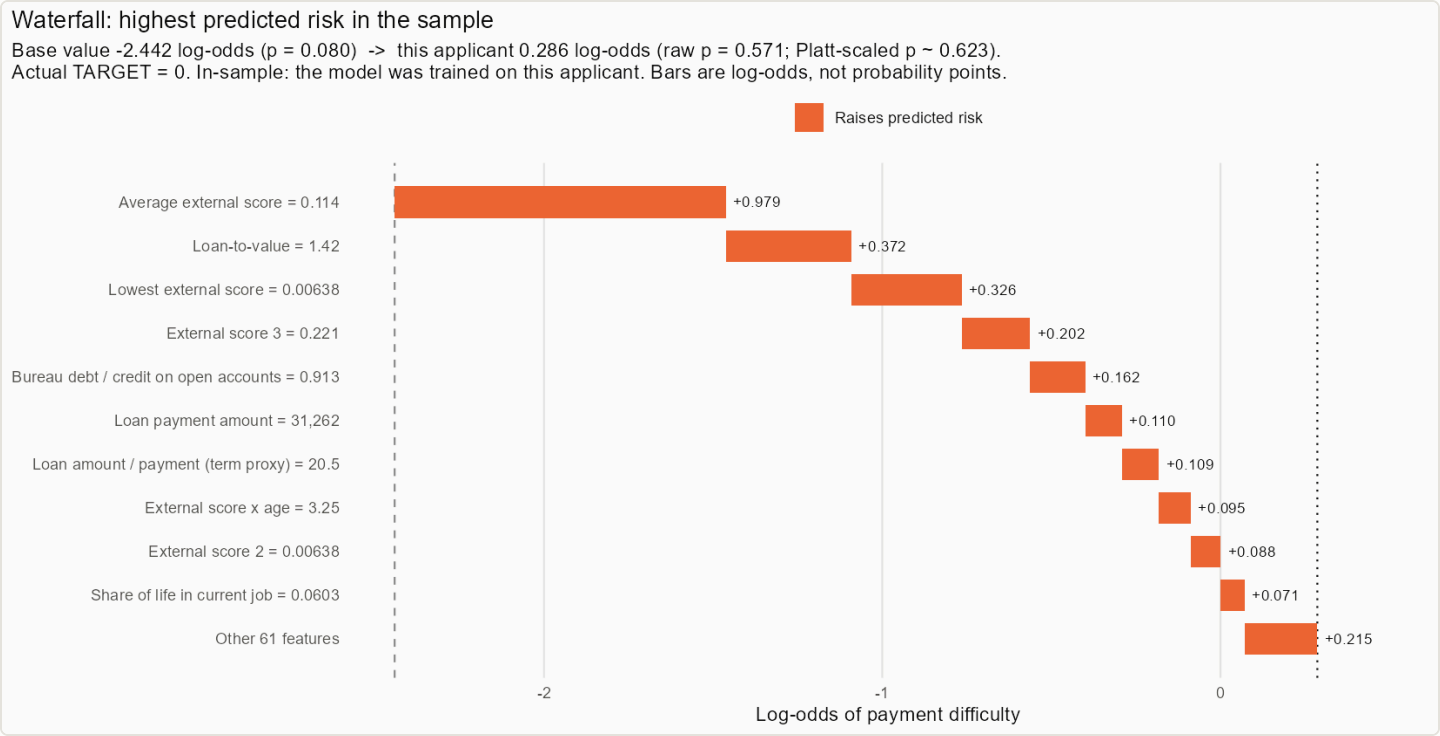


Figure 8: Highest-risk applicant in the sample (raw p = 0.571). Every top factor raises risk: very low external scores, loan-to-value of 1.42, and heavily used existing credit. This applicant did not actually have payment difficulty.

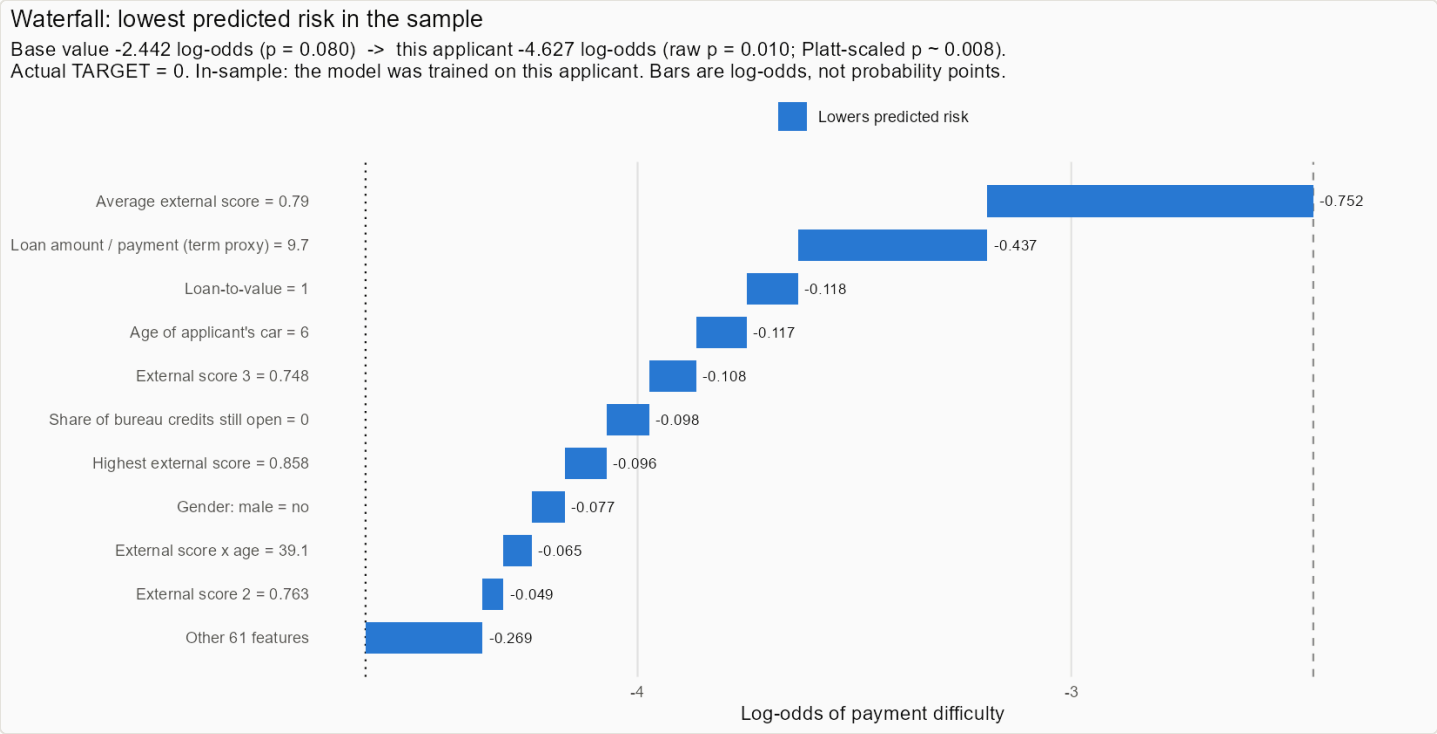


Figure 9: Lowest-risk applicant (raw p = 0.010): high external scores, a short loan, loan-to-value of 1.0 and no open bureau debt. Being female also lowers this prediction, which is a prohibited basis in a U.S. context.

Two applicants can receive very different predictions because each starts from the same base risk (about 8%) and is pushed up or down by their own characteristics. The pushes add on the log-odds scale, so the same feature value can move one applicant's *probability* more than another's.

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7 Adverse Action Mapping

U.S. creditors must give declined applicants the **specific principal reasons** for the decision (12 CFR 1002.9(b)(2)). The CFPB has stated that a complex algorithm is no excuse (Circular 2022-03), and that checklist reasons are not enough when they do not match the real reasons (Circular 2023-03, which the CFPB site now marks as archived). An applicant's reasons must be the factors that actually raised *that applicant's* predicted risk, not the model's global feature ranking.

Table 12: Candidate adverse-action explanations for the most influential risk-raising features

Model feature	Category	Candidate explanation	Usable as a reason?	Limitation / concern
`ext_source_mean / ext_source_min / ext_source_max / EXT_SOURCE_1–3 (+ missing flags, ext_mean_x_age)`	Undocumented external score	None can be stated accurately without vendor documentation (e.g., NOT 'poor credit score' or 'insufficient credit history').	No - not until the source, inputs and meaning of each score are documented	Largest driver of the model. CFPB Circ. 2022-03/2023-03: reasons must be specific and accurate; an unexplained third-party score cannot be translated by guesswork. ext_mean_x_age also embeds age.
`ltv`	Documented	Amount financed is high relative to the value of the goods purchased.	Yes, with validation	Mostly monotonic in this sample (Spearman 0.90); cite only when this applicant's ltv contribution is materially positive.
`credit_to_annuity`	Needs verification	Requested loan term / repayment schedule.	Partly - needs validation; it is a proxy, not the contractual term	Proxy for term; interaction-dependent. Explanation must be checked against the actual product terms.
`AMT_ANNUITY / AMT_GOODS_PRICE / AMT_CREDIT`	Documented	Amount of credit or payment requested.	Yes, with validation	Effects depend on other loan features; pair with affordability where relevant.
`bureau_debt_to_credit`	Documented	Outstanding balances are high relative to credit amounts on your existing accounts.	Yes, with validation	Imputed with the median when no bureau record - must not be cited for applicants without bureau data.
`bureau_active_share / bureau_active_credits`	Documented	Number of open credit accounts / proportion of accounts still open.	Yes, with validation	Imputed/zero when no bureau record; not applicable then.
`bureau_max_overdue_amount`	Documented	Delinquency reported on past or present credit obligations.	Yes	Standard, well-understood reason; verify data accuracy (FCRA-style disputes in a US context).
`employed_share_of_life / years_employed`	Needs verification (uses age)	Length of employment.	years_employed: Yes. employed_share_of_life: needs legal review	employed_share_of_life uses age in its denominator, so it is partly an age variable (Reg B 1002.6(b)(2)).
`OWN_CAR_AGE / FLAG_OWN_CARY`	Needs verification	Assets / collateral information.	Partly - needs business justification	Proxy for wealth; possible disparate impact; ownership rarely a direct repayment factor for unsecured loans.
`FLAG_DOCUMENT_3`	Undocumented field	None until the document is identified.	No	Unknown meaning; cannot produce a specific, accurate reason.
`years_since_id_publish`	Needs verification	Recent change of identification document.	Questionable - needs legal review	Not intuitively related to repayment; may proxy for life events or national origin/immigration status.
`DEF_30_CNT_SOCIAL_CIRCLE`	Legally problematic (third-party data)	None appropriate - reflects other people's behavior.	No	Data about third parties; CFPB Circ. 2023-03 highlights data not typically in a credit file; high fair-lending and reputational risk.
`CODE_GENDER (CODE_GENDERM)`	Prohibited basis	Must never be given - sex is a prohibited basis.	No - prohibited basis	12 CFR 1002.6(b)(9): a creditor shall not consider sex in any aspect of a credit transaction. Must be removed before any US-style use.

Model feature	Category	Candidate explanation	Usable as a reason?	Limitation / concern
`NAME_EDUCATION_TYPE`	Legally problematic (disparate-impact risk)	Should not be used as an adverse-action reason without legal review.	No - not without legal and business justification	Not a listed prohibited basis, but strongly correlated with protected characteristics; disparate-impact risk.
`NAME_FAMILY_STATUS`	Prohibited basis	Must never be given - marital status is a prohibited basis.	No - prohibited basis	12 CFR 1002.2(z) and 1002.6(b)(8): married and unmarried applicants must be evaluated by the same standards.
`age_years / age_band`	Restricted (age)	Only if the system qualifies as empirically derived and sound and elderly applicants are never disfavored; otherwise none.	No - restricted (needs legal review)	12 CFR 1002.6(b)(2): age may be used only in an empirically derived, demonstrably and statistically sound system, and elderly (62+) applicants must not receive a negative value. Checked in step6_elderly_check.csv.

The model cannot provide reliable adverse-action reasons as built. Among sampled applicants who would be declined at 0.25:

- for **98%**, the main risk-raising factor is an undocumented external score;
- for **42%**, gender is among the top four factors.

Usable reasons are limited to documented loan-structure, bureau-debt and employment features, and even those need per-applicant validation. External scores could only be translated into plain-English reasons after the provider documents what they measure. They must not be relabeled as “poor credit score” or “insufficient credit history”. Gender, marital status, education, age and social-circle data must never be given as reasons.

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8 Fairness Analysis

All results below are on the untouched holdout at the frozen 0.25 threshold. **Statistical disparities do not by themselves prove unlawful discrimination, and they do not rule it out.** That requires legal analysis.

Table 13: Holdout outcomes by gender and education (95% intervals in brackets)

Attribute	Group	n (events)	Approval	Observed payment difficulty	False positive rate	True positive rate	Precision	AUC	Predicted / observed	Reliability
Gender	F	40,476 (2,854)	96.3% [96.1%, 96.4%]	7.05%	2.7%	17.2%	32.4%	0.749 [0.739, 0.758]	7.03% / 7.05%	Adequate
Gender	M	21,026 (2,111)	92.4% [92.0%, 92.7%]	10.04%	5.6%	26.4%	34.7%	0.752 [0.742, 0.763]	9.86% / 10.04%	Adequate
Education	Higher education	15,099 (799)	98.1% [97.9%, 98.3%]	5.29%	1.4%	9.8%	27.9%	0.751 [0.733, 0.769]	5.55% / 5.29%	Adequate
Education	Secondary / secondary special	43,544 (3,910)	93.9% [93.7%, 94.1%]	8.98%	4.4%	23.2%	34.1%	0.749 [0.741, 0.757]	8.76% / 8.98%	Adequate
Education	Incomplete higher	2,066 (180)	93.9% [92.8%, 94.9%]	8.71%	4.3%	24.4%	35.2%	0.749 [0.711, 0.787]	9.25% / 8.71%	Adequate
Education	Lower secondary ⚠	756 (75)	93.0% [90.9%, 94.6%]	9.92%	5.3%	22.7%	32.1%	0.728 [0.665, 0.791]	9.70% / 9.92%	Limited (small group)
Education	Academic degree ⚠	37 (1)	94.6% [82.3%, 98.5%]	2.70%	—	—	—	—	—	Unreliable (too few applicants or events)

⚠ **Academic degree** (37 applicants, 1 payment-difficulty case) is too small to support any conclusion. **Lower secondary** (756 applicants, 75 cases) has wide intervals.

Table 14: Key disparities (group minus reference; applicant-level bootstrap 95% CI)

Comparison	Approval gap	Approval ratio	False positive rate gap	True positive rate gap	Payment-difficulty gap
M vs F	-3.9 pp [-4.3 pp, -3.5 pp]	0.959	+2.8 pp [+2.5 pp, +3.2 pp]	+9.2 pp [+6.7 pp, +11.6 pp]	+3.0 pp [+2.5 pp, +3.5 pp]
Secondary / secondary special vs Higher education	-4.3 pp [-4.6 pp, -3.9 pp]	0.957	+3.0 pp [+2.7 pp, +3.3 pp]	+13.5 pp [+11.1 pp, +15.9 pp]	+3.7 pp [+3.2 pp, +4.1 pp]
Single / not married vs Married	-3.2 pp [-3.8 pp, -2.6 pp]	0.966	+2.3 pp [+1.8 pp, +2.9 pp]	+7.0 pp [+3.8 pp, +10.1 pp]	+2.4 pp [+1.7 pp, +3.1 pp]
< 30 vs 40-49	-6.6 pp [-7.3 pp, -5.8 pp]	0.931	+5.2 pp [+4.6 pp, +5.9 pp]	+9.7 pp [+6.0 pp, +13.4 pp]	+4.6 pp [+3.8 pp, +5.4 pp]

Gender.

- **Approval:** women 96.3%, men 92.4%, a gap of **3.9 percentage points**.
- **Repayers declined:** men who would *not* have had payment difficulty are declined at 5.6%, against 2.7% for women, about twice as often.

- **Ranking and calibration:** the same for both groups (AUC 0.749 vs 0.752).

Education.

- **Approval:** higher education **98.1%**, secondary 93.9%, a gap of **about 4.3 points**.
- **Repayers declined:** secondary-educated repayers are declined 3.1× as often (4.4% vs 1.4%).

The 98.1% figure is corrected from an earlier 98.2% that came from rounding twice.

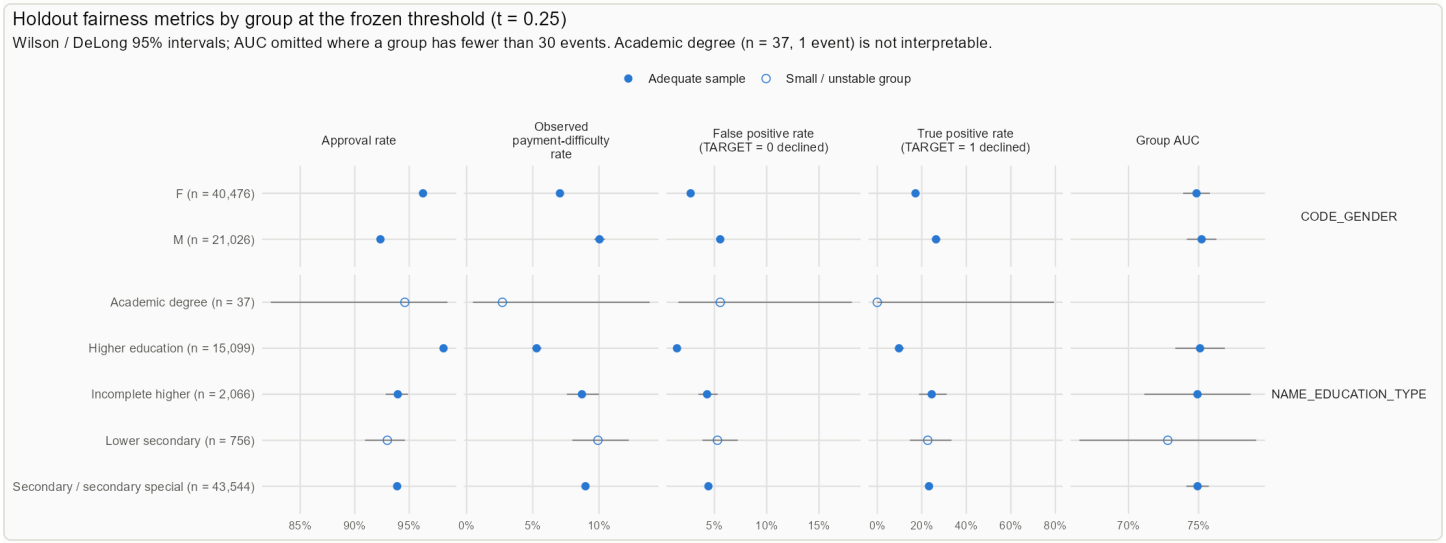


Figure 10: Holdout fairness metrics by group at t = 0.25, with 95% intervals. Hollow points are small or unstable groups.

Which fairness criteria are met?

Table 15: Fairness criteria at the 0.25 threshold

Criterion	Meaning here	Result
Demographic parity	Equal approval rates across groups	**Not met.** Gaps of 3.9 points (gender) and 4.3 points (education); approval ratios of about 0.96, above the four-fifths heuristic from U.S. employment guidance, which is not a lending legal test
Equal opportunity	Applicants who would repay are approved at equal rates (equal false positive rates)	**Not met.** 2.0× (gender) and 3.1× (education)
Equalized odds	Equal false positive *and* true positive rates	**Not met**
Predictive calibration	A given score means the same risk in every group	**Approximately met** for adequately sized groups; weaker for applicants aged 62+ (AUC 0.686, slope 0.78)
Ranking quality	Equal AUC across groups	**Met** (no significant differences)

When observed rates differ between groups (men 10.0% vs women 7.1%; secondary 9.0% vs higher education 5.3%), a calibrated score with one threshold generally cannot also equalize error rates (Chouldechova, 2017; Kleinberg et al., 2016). The rate differences explain part of the outcome gaps but do not justify them automatically. The labels measure early payment difficulty rather than loss, outcomes exist only for previously approved applicants, and much of the gender gap comes from the model using gender directly.

Direct use of protected characteristics, and proxies. Holdout SHAP values from the development-only model were used for a descriptive accounting: remove one attribute's term and re-apply the threshold. This is not a retrained model.

Table 16: Approval gaps with and without the model's direct use of each attribute (holdout)

Gap	Direct term removed	Actual	Without the term	Share explained by the direct term
Approval gap F - M	Gender (CODE_GENDER)	+3.9 pp	+1.4 pp	64%
Approval gap Higher - Secondary	Gender (CODE_GENDER)	+4.3 pp	+4.0 pp	7%
Approval gap Married - Single	Gender (CODE_GENDER)	+3.2 pp	+2.8 pp	13%
Approval gap F - M	Marital status (NAME_FAMILY_STATUS)	+3.9 pp	+3.8 pp	2%
Approval gap Higher - Secondary	Marital status (NAME_FAMILY_STATUS)	+4.3 pp	+4.2 pp	2%
Approval gap Married - Single	Marital status (NAME_FAMILY_STATUS)	+3.2 pp	+2.7 pp	17%
Approval gap F - M	Education (NAME_EDUCATION_TYPE)	+3.9 pp	+3.7 pp	5%
Approval gap Higher - Secondary	Education (NAME_EDUCATION_TYPE)	+4.3 pp	+2.5 pp	42%
Approval gap Married - Single	Education (NAME_EDUCATION_TYPE)	+3.2 pp	+3.3 pp	-5%
Approval gap F - M	All of the above	+3.9 pp	+1.1 pp	73%
Approval gap Higher - Secondary	All of the above	+4.3 pp	+1.9 pp	54%
Approval gap Married - Single	All of the above	+3.2 pp	+2.5 pp	21%

- **Gender:** a direct input that raises calibrated risk for **every** man (about +1.05 percentage points on average). It accounts for about **64%** of the gender approval gap.
- **Education:** a direct input. Higher education lowers risk by about 1.0 point, and the direct term accounts for about 42% of the education gap.
- **Marital status:** a direct input. Unmarried categories raise risk by about 0.2 points.
- **Age:** a direct input, and also part of two engineered features. **197 of 4,724 applicants aged 62+** received a risk-raising direct age contribution. U.S. Regulation B allows age in scoring only if elderly applicants are never assigned a negative value.
- **Proxies:** the gap that remains after removing the direct terms (about 1.1 points for gender) runs through other features. That can reflect genuine risk differences, proxies, or both. No single influential feature strongly stands in for gender or education; car ownership is the strongest (moderate). Combinations of features have not been tested.

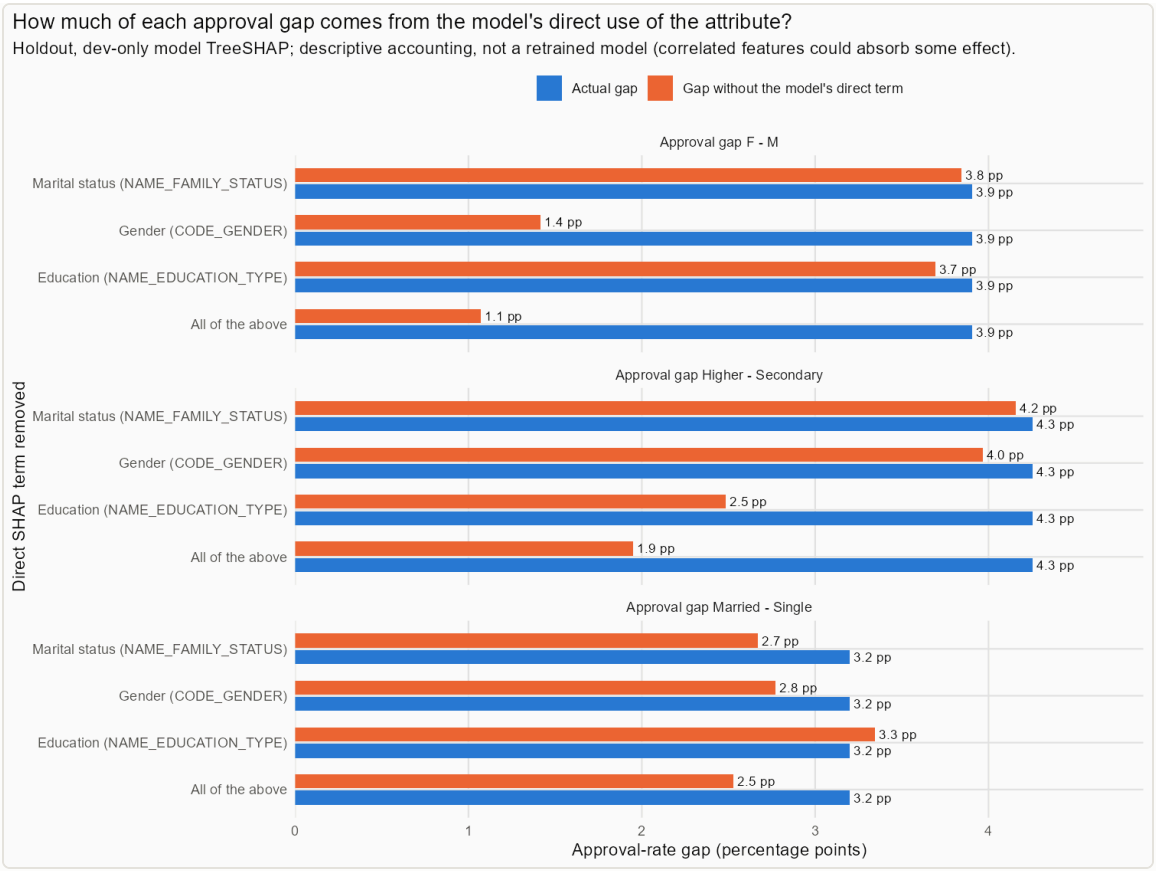


Figure 11: How much of each approval gap comes from the model's direct use of an attribute (holdout; descriptive SHAP accounting).

Threshold sensitivity (descriptive only; the threshold stays 0.25).

Table 17: Group disparities at alternative thresholds (holdout; sensitivity analysis only)

Threshold	Overall approval	Gender approval gap	Gender FPR ratio (M/F)	Education approval gap	Education FPR ratio (Sec./Higher)
0.15	85.3%	9.1 pp	1.83×	10.0 pp	2.38×
0.20	91.2%	6.2 pp	1.98×	6.8 pp	2.80×
0.25 (frozen)	94.9%	3.9 pp	2.04×	4.3 pp	3.13×
0.30	97.2%	2.4 pp	2.20×	2.5 pp	3.28×
0.35	98.4%	1.4 pp	2.23×	1.4 pp	3.41×

Raising the threshold narrows the absolute approval gaps, mostly because nearly everyone gets approved, but it **widens** the relative false-positive disparity. No threshold in this range removes the disparities, and changing the threshold does not address the direct use of protected inputs.

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9 Limitations and Deployment Risks

Rating criteria:

- **Critical:** a prohibited or restricted practice, or something that prevents compliant operation. Blocks deployment.

- **High:** material uncertainty that could change who is approved, the economics, or customer harm. Must be resolved before a pilot.
- **Medium:** manageable with standard validation, monitoring and governance.
- **Low:** limited impact; document and monitor.

Table 18: Deployment risk register

Risk	Rating	Evidence	Required mitigation or validation
Direct use of gender and marital status	**Critical**	CODE_GENDER is the #4 variable by SHAP; it raises predicted risk for every male applicant. Marital status is also a direct input. Both are prohibited bases under 12 CFR 1002.2(z) / 1002.6(b).	Remove both; rebuild and re-validate a candidate model; legal sign-off before any use.
Age-related restrictions	**Critical**	age_years, age_band, ext_mean_x_age and employed_share_of_life use age; some elderly (62+) applicants receive a risk-raising age contribution, which 12 CFR 1002.6(b)(2)(ii) does not allow.	Remove age or constrain it (e.g., monotone so elderly are never disfavored); legal review of age-derived features.
Unexplained external credit scores	**Critical**	EXT_SOURCE family is ~half of model attribution and the main reason for ~98% of would-be declines; provider, inputs and meaning are undocumented.	Obtain vendor documentation, permissible-purpose and data-accuracy assurances; otherwise exclude.
Missing valid adverse-action explanations	**Critical**	Principal reasons for most declines are undocumented scores or prohibited bases; generic checklist reasons are insufficient (CFPB Circ. 2022-03, 2023-03; 12 CFR 1002.9(b)(2)).	Build and test a reason-code process on the compliant model; legal and customer-communication review.
Education disparities	**High**	Education is the #5 variable; Higher-education approval 98.1% vs Secondary 93.9%; FPR 1.4% vs 4.4%.	Document business justification; test less-discriminatory alternatives; monitor outcomes by education.
Proxy discrimination	**High**	Several influential features differ by gender/education (see step7_proxy_screen.csv); removing direct terms will not remove all group differences.	Proxy analysis and disparate-impact testing on the compliant model; review car ownership, income type, ID recency, social-circle data.
Labels are payment difficulty, not losses	**High**	TARGET = 1 is late payment within the first Y installments; X and Y undisclosed; loss conversion c unsourced.	Re-label with observed charge-off and recovery outcomes; validate c, LGD, EAD.
Loan-term and financial-assumption uncertainty	**High**	Base-case gain +$15.8/applicant at 24 months but ~+$0.8 (CI spans zero) at 36 months; optimal threshold 0.11-0.48 across plausible assumptions.	Use Home Credit's actual pricing, term, funding and cost data; re-derive threshold before any pilot.
Selection bias (outcomes only for approved applicants)	**High**	Training data contain past approvals only; performance on previously declined applicants is unknown.	Reject-inference study or controlled pilot with limited exposure; monitor approval-population shift.
Geographic and temporal generalization	**High**	Country, currency and period of the data are undisclosed; Home Credit operated in 9 countries; economics used U.S. and 2018 Home Credit figures.	Out-of-time and per-market validation; re-calibrate per market.
Model calibration and generalization	**Medium**	Holdout AUC 0.755 vs CV 0.759; Platt slope 0.99 on holdout; calibration similar across gender and education groups.	Re-confirm on out-of-time data; monitor calibration and AUC monthly; recalibration trigger rules.
Monitoring and governance	**Medium**	No monitoring, drift detection, override process or model-risk sign-off exists yet.	Model-risk-management review (independent validation), drift/fairness dashboards, human-review process for referrals, documented change control.
Small-group instability	**Low**	Academic degree n = 37 (1 event); Lower secondary n = 756; results not interpretable or limited.	Pool or report with uncertainty; collect more data before conclusions.

Further limitations:

- **Label vs loss.** TARGET = 1 is early payment difficulty with undisclosed definitions (X days, first Y installments). It is not charge-off, recovery or loss.
- **Financial assumptions.** The loss conversion (*c*) and the base-case loss given default are assumptions. The loan term is unresolved (24 vs 36 months). Margin figures are 2018 portfolio averages across nine countries.
- **Selection bias.** Outcomes exist only for previously approved applicants, so performance on applicants the lender used to decline is unknown.
- **Domain and time.** The country, currency and period are undisclosed. Results may not carry over to other markets or later periods, and no out-of-time test was possible.
- **Historical feature-selection leakage.** Some EDA bands and choices were made by looking at all labeled rows, holdout included. Fold-specific re-preprocessing showed the learned parameters made a negligible difference (AUC +0.00002), but the EDA-derived choices could not be refit.
- **Explanations.** SHAP comes from the full-data model on applicants it was trained on. Credit is split among correlated features, especially the external-score summaries and the age interactions.
- **Small groups.** Academic degree (37 applicants) and lower secondary (756) results are not reliable.
- **Jurisdiction.** U.S. ECOA/Regulation B is the reference framework, but the data's jurisdiction is unknown. Nothing here is legal advice.

Conditions before deployment could be considered:

1. **Legal review** of sex, marital status, age and age-derived features, education, income type, car ownership, social-circle data and ID-document recency.
2. **A compliant candidate model:**
   - prohibited inputs removed;
   - age removed, or constrained so elderly applicants are never disfavored;
   - validated with the same development/holdout/calibration discipline;
   - plus proxy, disparate-impact and less-discriminatory-alternative testing.
3. **Documentation of the external scores** (provider, inputs, permissible use, accuracy), or their exclusion, plus specific, tested adverse-action reason codes.
4. **Real lending economics:** observed charge-offs, recoveries, terms, pricing and costs, with the threshold re-derived from them.
5. **Independent model validation**, out-of-time and per-market testing, a reject-inference study or limited pilot, fairness and drift monitoring, and human review of referred applicants.

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- 12 CFR 1002.6 (rules concerning evaluation of applications, including sex, marital status and age). https://www.consumerfinance.gov/rules-policy/regulations/1002/6/
- 12 CFR 1002.9 (notifications; statement of specific reasons). https://www.consumerfinance.gov/rules-policy/regulations/1002/9/
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Our own assumptions (not published findings)

- Loss conversion c = 25% / 50% / 100% of TARGET = 1 cases becoming charge-offs (no source exists).
- Base-case loss given default of 70.5%: the midpoint of the sourced 45% and 96% endpoints.
- Installments paid before default: 0 / 3 / 6.
- Loan term of 24 months at 40% APR, derived from previous applications in the dataset (36 months shown as an alternative).
- Illustrative principal of $8,437.
- Cured late payers earn zero; TARGET = 0 loans are fully repaid.
- The 0.25 threshold: a research setting, not a lending policy.